more than 20 billion data points annually, with dashboards monitoring 300+ process parameters in real time.

We are actively exploring Generative AI for engineering and quality root-cause analysis, industrial digital twins for scenario planning, and 5G integration for ultra-reliable low-latency communication. Collaboration across our global smart factory network, including Hyundai Motor Group Innovation Centre Singapore (HMGICS), accelerates learning cycles and ensures responsible scaling through shared standards in HMIL, cyber-physical security and AI governance.

THE NEXT ERA OF MANUFACTURING IS ALREADY HERE. **PREDICTIVE ANALYTICS, DIGITAL TWINS, AUTONOMOUS ROBOTS** AND SOFTWARE-INTEGRATED OPERATIONS ARE CONVERGING ON SHOP FLOORS

The industry as a whole is moving toward dark factories—high-automation environments where AI systems manage inspection, assembly and intralogistics, with minimal human presence. Digital tools expand human judgement and creativity, elevating roles from repetitive tasks to higher-order problem-solving and orchestration. Humanoid and collaborative robots (cobots) will increasingly share space with frontline teams, particularly in ergonomically challenging or high-precision tasks. Automation delivers consistency; humans provide adaptability and innovation. The result is a shop floor that is safer, more predictable and more resilient to demand fluctuations.



Illustration by **NILANJAN DAS / AI**